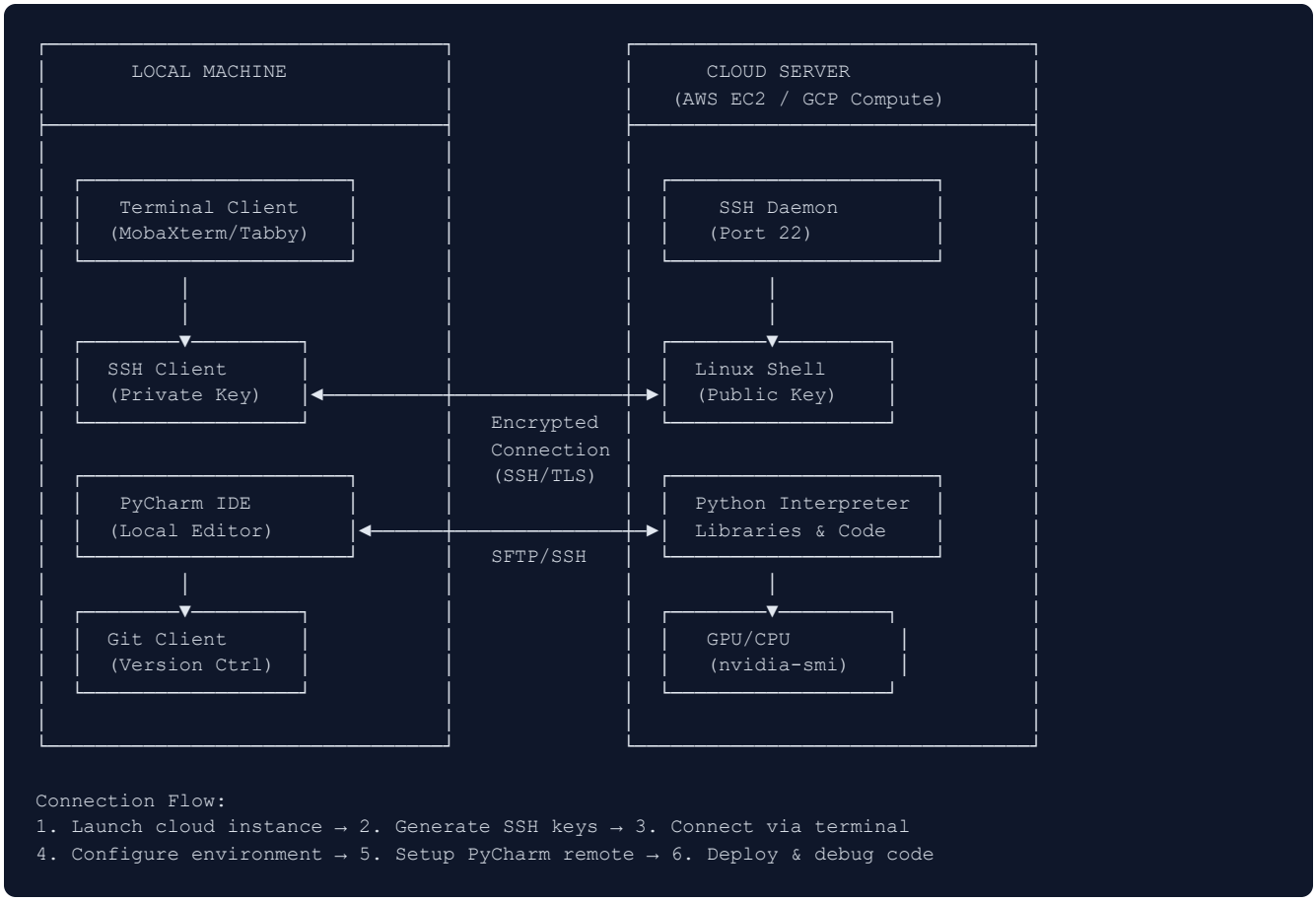
Cloud Computing Software Setup Checklist

A comprehensive guide to setting up your development environment for cloud computing coursework.

Getting Started

- ☐ **Introduction** - Remote connections, SSH, terminals, cloud platforms, servers, VMs, engines, and AMI concepts

Architecture Overview



Cloud Platform Fundamentals

- ☐ **What is a Dashboard?** - Navigate cloud provider interfaces
- ☐ **Going Over the Document: How to Make an EC2 Instance with AWS** - Launch and configure EC2 instances

- Change instance type - Modify CPU/RAM resources
- Stop instances - Pause instances to save costs
- Terminate instances - Permanently delete instances
- IP address changes - Handle dynamic/static IP addresses
- Elastic IP - Reserve static IP addresses
- Security groups - Configure firewall rules
- Key pairs - Manage SSH authentication

Terminal Setup

- [ ] **Terminal (MobaXterm, Tabby)** - Install and configure terminal emulator
- [ ] **Useful Terminal Commands** - Master essential Linux commands
- `ls` - List directory contents
- `cd` - Change directory
- `pwd` - Print working directory
- `mkdir` - Create directory
- `rm` - Remove files/directories
- `cp` - Copy files/directories
- `mv` - Move/rename files
- `chmod` - Change file permissions
- `chown` - Change file ownership
- `nano` / `vim` - Text editors
- `top` - Process monitoring
- `htop` - Interactive process viewer
- `nvidia-smi` - NVIDIA GPU monitoring
- `nvtop` - Interactive GPU process viewer
- `screen` - Keep processes running after disconnect
  - `screen -S mywork` - Start new session named "mywork"
  - `Ctrl+A, D` - Detach from session (keeps running)
  - `screen -ls` - List all screen sessions
  - `screen -r mywork` - Reattach to session "mywork"
  - `exit` - Close current screen session

SSH Configuration

- [ ] **SSHing (AWS Mac and Windows)** - Establish secure shell connections

File Transfer (SFTP)

- [ ] **SFTP File Management** - Transfer and browse remote files securely
- MobaXterm - Built-in SFTP browser (left sidebar)
- FileZilla - Standalone SFTP client for file transfers
- PyCharm - Integrated remote file browser and sync

PyCharm IDE Setup

- [ ] **PyCharm Setup** - Install and configure PyCharm Professional
- [ ] **PyCharm Folder Mapping** - Sync local and remote directories
- [ ] **PyCharm Remote Interpreter** - Execute code on remote Python environment
- [ ] **PyCharm SSH** - Connect PyCharm to remote server
- [ ] **PyCharm SSH Debugging** - Debug code on remote server
- [ ] **PyCharm Git** - Version control integration
- [ ] **PyCharm Projects and Coding** - Develop and test applications

Google Cloud Platform (GCP)

- [ ] **GCP Coupons** - Activate free credits
- [ ] **GCP SSH Key Generator** - Create authentication keys
- [ ] **GCP Compute Engine** - Launch GCP virtual machines
- [ ] **GCP Cloud Storage** - Use cloud object storage
- [ ] **GCP and PyCharm** - Integrate GCP with PyCharm
- [ ] **GCP and PyCharm SSH** - Connect PyCharm to GCP instances

S3

- [ ] **S3 Bucket Creation** - Set up and manage S3 buckets
- [ ] **S3 File Upload** - Transfer files to S3
- [ ] **S3 File Download** - Retrieve files from S3
- [ ] **S3 Bucket Policies** - Configure access permissions
- [ ] **S3 Versioning** - Enable versioning for data protection

Troubleshooting & Advanced Topics

- [ ] **Change IP Each Time the Instance Is Restarted** - Handle dynamic vs static IPs
- [ ] **Change Instance Type** - Modify compute resources

Completion Notes

Once completed, you should have functional cloud instances on AWS/GCP, working SSH connections, PyCharm configured for remote development, and the ability to deploy and manage cloud-based applications.